

CSCI 5253

Datacenter Scale Computing - Methods, Systems and Techniques

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	Machine Learning Engineer (71.8 / 100)	62.5 / 100
SPECIALIZATION On-Campus Elective		

62

CURRENT AVG SCORE
across 6 job roles

93

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Covers the primary problem solving strategies, methods and tools needed for data-intensive programs using large collections of computers typically called "warehouse scale" or "data-center scale" computers. Examines methods and algorithms for processing data-intensive applications, methods for deploying and managing large collections of computers in an on-demand infrastructure and issues of large-scale computer system design.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	71.8	100.0	+28.2	MEDIUM
AI Engineer	69.9	97.5	+27.6	MEDIUM
ML Engineer	63.2	85.9	+22.7	MEDIUM
MLOps Engineer	60.5	75.9	+15.4	MEDIUM
Applied Scientist	60.1	100.0	+39.9	HIGH
Data Scientist	49.4	98.0	+48.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

- 2 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

- 3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

71.8 → 100.0 (+28.2 pts)

data web engineering sets learning analytics business mobile

AI Engineer

69.9 → 97.5 (+27.6 pts)

data robotics engineering models learning build software systems

ML Engineer

63.2 → 85.9 (+22.7 pts)

media discovery data learning machine systems models engineering

MLOps Engineer

60.5 → 75.9 (+15.4 pts)

detection data systems learning machine preprocessing media image

Applied Scientist

60.1 → 100.0 (+39.9 pts)

discovery scientific research reasoning problems models data learning

Data Scientist

49.4 → 98.0 (+48.5 pts)

data sets analytics evolution social build engineering people

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.